Exploratory Data Analysis (EDA) Summary
Report Template

1. Introduction

The purpose of this Exploratory Data Analysis (EDA) is to examine the **Delinquency Prediction Dataset** before building a predictive model. The analysis focuses on understanding the dataset structure, identifying missing values and data quality issues, detecting patterns and anomalies, and discovering the key risk factors associated with customer delinquency. These findings will help improve data quality and support the development of an accurate delinquency prediction model.

2. Dataset Overview

This dataset contains customer financial and credit-related information used to predict whether a customer is likely to become delinquent.

Key Dataset Attributes

- **Number of records:** 500
- **Number of variables:** 19
- **Target variable:** Delinquent_Account (0 = Non-Delinquent, 1 = Delinquent)

Variable	Description
Customer_ID	Unique customer identifier
Age	Customer age
Income	Annual income
Employment_Status	Employment category
Credit_Score	Customer credit score
Loan_Balance	Outstanding loan amount

Variable	Description
Credit_Utilization	Percentage of available credit used
Debt_to_Income_Ratio	Debt compared to income
Missed_Payments	Number of previous missed payments
Month_1 – Month_6	Monthly payment history
Delinquent_Account	Target variable

Data Types

- Numerical Variables
  - Age
  - Income
  - Credit Score
  - Loan Balance
  - Credit Utilization
  - Debt-to-Income Ratio
  - Missed Payments
- Categorical Variables
  - Employment Status
  - Credit Card Type
  - Location
  - Monthly Payment Status

Initial Observations

- No duplicate records were found.
- Missing values exist in Income, Loan Balance, and Credit Score.
- Employment Status contains inconsistent category names.
- Credit Utilization contains values greater than 100%, which require validation.

3. Missing Data Analysis

Missing data was identified in several important financial variables. Since the percentage of missing values is relatively small, imputation was selected instead of deleting records.

Variables with Missing Values

Variable	Missing Values	Treatment
Income	39	Median Imputation
Loan_Balance	29	Median Imputation
Credit_Score	2	Mean Imputation

Additional Data Quality Issues

Issue	Treatment
Employment_Status contains "Employed", "employed", "EMP"	Standardized into one category
Credit_Utilization values greater than 100%	Validated and capped/corrected

Justification

- Median imputation was selected for Income and Loan Balance because financial variables often contain outliers.
- Mean imputation was sufficient for Credit Score because only two values were missing.
- Standardizing categorical values improves model consistency.
- Correcting invalid utilization values prevents misleading model predictions.

4. Key Findings and Risk Indicators

Correlations Observed Between Key Variables

- High Credit Utilization is positively associated with delinquency risk.
- Customers with low Credit Scores are more likely to become delinquent.

- Previous Missed Payments strongly predict future delinquency.
- Higher Debt-to-Income Ratios increase repayment risk.
- Large Loan Balances may contribute to higher default probability.
- Unstable Employment Status may increase financial risk.

Unexpected Anomalies

- Employment Status contains inconsistent category labels.
- Credit Utilization includes values exceeding 100%.
- Target variable is imbalanced, with approximately **16% delinquent** and **84% non-delinquent** accounts

5. AI & GenAI Usage

Generative AI (ChatGPT) was used to summarize the dataset, identify missing values, recommend appropriate imputation techniques, detect anomalies, and identify key delinquency risk factors.

Example AI Prompts Used

- "Summarize key patterns, outliers, and missing values in this dataset."
- "Identify the top three variables most likely to predict delinquency."
- "Suggest an imputation strategy for missing values based on industry best practices."
- "Detect early indicators of delinquency risk in the dataset."
- "Recommend methods to handle missing financial variables for predictive modelling."

AI-Assisted Insights

- Recommended median imputation for financial variables.
- Identified Credit Utilization, Credit Score, and Missed Payments as the strongest predictors.
- Highlighted inconsistent Employment Status categories.
- Identified anomalous Credit Utilization values above 100%.
- Recommended handling class imbalance during model building.

6. Conclusion & Next Steps

The exploratory analysis indicates that the dataset is suitable for predictive modelling after minor preprocessing. Missing values were handled using appropriate imputation methods, inconsistent categorical values were standardized, and invalid Credit Utilization values were identified for correction. The most influential predictors of delinquency are **Credit Utilization, Credit Score, Missed Payments, Debt-to-Income Ratio,** and **Loan Balance**.

Recommended Next Steps

- Complete data preprocessing and feature engineering.
- Encode categorical variables.
- Normalize or scale numerical features where required.
- Address class imbalance using techniques such as SMOTE or class weighting.
- Train and evaluate machine learning models (e.g., Logistic Regression, Decision Tree, Random Forest, XGBoost).
- Compare model performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC before selecting the final model.